Portfolio.

Yang Kang
Product Designer

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(form a to z)

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Me.
(in short)

0.

Yang Kang industrial designer.

"Hi! I'm a product designer currently based in Shenzhen. Over the past 5 years, I've built a career around one core philosophy: design is a logical system, not just an aesthetic choice. I'm passionate about building products that work as well as they look. I'm currently looking for my next challenge if you're looking for a design partner who values a problem-solving mindset and innovation, let's get in touch."

+86 17673205160

Yueyang Huhan China

ykbnn125@163.com

921100579/17673205160



Education&Experiences.

(More about me)

Educational path

Brunel University London

Master's degree **Product design** 2019.8-2020.11

Hunan university of science and technology 2015.6-2019.3

Bachelor's degree **Product design**

Collaborations and work experiences

Hunan Tuoerqi Network Technology Co., Ltd.

Chang Sha
2025.4-2025.11

The company specializes in webcams for the U.S. market, with Amazon as its primary sales channel. During my six months there, I was responsible for the design of five products that were successfully brought to market.

usmile Kitty Annie 小猫爱旅

Shen Zhen
2021.1-2022.5

In my first formal role, I was responsible for the entire design research process, turning user insights into actionable concepts. and worked closely with factories.

Firstuniôn

Shen Zhen
2022.5-2024.12

Expertise in consumer electronics I managed the full workflow from initial design to lockdown, successfully launching several products currently on the market.

QUICK-E

London
2020.1-2020.9

A company that makes power bank located at London, I working as a design intern. my job is creating high-quality imagery and brainstorming innovative concepts for power bank solutions.

Skills.

(An overview of my capabilities)

Soft skills

- Problem solving
- Creative
- Curious
- Teamwork
- Adaptable
- CMF processing
- Concentrate
- Resilient
- Communication
- Fast learner

Native speaker

IELTS:6.5

Software and tools

Rhino

KeyShot

Photoshop

Illustrator

cero

Solidthinking
Evolve

Tiny But ‘BIG’

(Aim to design a second toothbrush for electronic toothbrush user)

01
toothbrush

The second toothbrush for electronic toothbrush user.

1. Vision & Strategy
I acted as both Product Manager and Designer. The first step was to identify whether there was a market gap for a high-performance, portable toothbrush.

2. Target Users
Specifically designed for frequent business travelers and younger consumers (Gen Z), every feature supports their need for efficiency, portability, and a modern, youthful aesthetic.

3. Strategic Material Design
To balance sophistication with budget constraints, a unique “speckled” eco-composite material was selected. This reduced production costs while delivering a premium, textured appearance that stands out on the shelf.

4. The Result
The result is a market-ready, cost-effective product that demonstrates a thoughtful material choices and a strong understanding of user needs can create a thoughtful design.



Tiny But ‘BIG’

(Aim to design a second toothbrush for electronic toothbrush user)

01
toothbrush

Benchmarking

At the beginning of the project, I carried out market research to confirm the strong demand for portable electric toothbrushes. Here are some of the key findings.

Key words: Portable toothbrushes.
A day's business records at taobao

Number of people searching	number of searches	ctr	Pay Conversion Rate
10,169	20,176	93.18%	19.21%





The result indicates substantial demand for portable solutions. The current conversion rate of 19.21% suggests a clear opportunity to introduce a more optimized, user-centric product to better serve this market.

Tiny But ‘BIG’

(Aim to design a second toothbrush for electronic toothbrush user)

01
toothbrush

Persona_

Who are the target users? Based on market research and data analysis, I defined two primary user groups.

Office woman



Typical user
weiyu,28,married
-who travel frequently.
-Pay attention to oral health
-High income.
-Tech-forward efficiency.

Office woman



Unfollowed users
liu yun, 21,single
-Budget-conscious
-18-25 female or 22-28 male
-Looking for a professional entry-point into electric oral care.

To appeal to both high-income frequent travelers and tech-curious younger users, the design needed to balance aesthetics with professional-level performance. The goal was to create a product that feels dynamic yet dependable.

Tiny But ‘BIG’

(Aim to design a second toothbrush for electronic toothbrush user)

01
toothbrush

Main rival



Brand: 3D cat glue
Price: 299
Monthly sales: 1245
Function: electric toothbrush with a box (5min sterilization, 25min circulating air drying)
pros:Small, can be air-dried and sterilized, easy to carry on business trips
cons:The brush head is a bit small, handle is too short, power run low too quick, and the air-drying sound is loud.



Brand: TFLASH
Price: 279-349
Monthly sales: 4354
Function: Wireless charging, Base storage, 2 in 1, 4 gear modes, UV, with a Cup
pros:Ultraviolet sterilization, easy to carry on business trips, low noise.
cons:The brush head is a bit small, handle is too short, power run low too quick.

Current portable options show limited innovation, often repackaging standard toothbrushes with simple protective covers. Key pain points include compromised ergonomics (shortened handles), excessive noise, and limited battery life that requires frequent charging.

Tiny But ‘BIG’

(Aim to design a second toothbrush for electronic toothbrush user)

01
toothbrush

Design keywords and Sum up

Driven by my research, I defined three core design keywords: Young, Professional, and Polychromatic.







Tiny But ‘BIG’

(Aim to design a second toothbrush for electronic toothbrush user)

01

toothbrush







Toothbrush/M

(Aim to design a toothbrush for male user group)

Toothbrush for male.

1. Market Opportunity
A gap was identified in the product lineup, which previously leaned toward more feminine aesthetics. Research was led to establish a new design language tailored specifically to male users.

2. Design Keywords
Defined three core pillars for the project: "Simple," "Texture," and "Technology." These keywords guided every decision, from the ideal to the final product.

3. Details on a Budget
Within tight cost constraints. A subtle breathing light around the power button enhances the tech-forward feel, while pearlescent paint creates a refined metallic effect without relying on expensive materials.

4. The Result
A minimalist shape while delivering a tech-forward experience for the user.



Toothbrush/M

(Aim to design a toothbrush for male user group)

02

toothbrush



Toothbrush/M

(Aim to design a toothbrush for male user group)

02

toothbrush





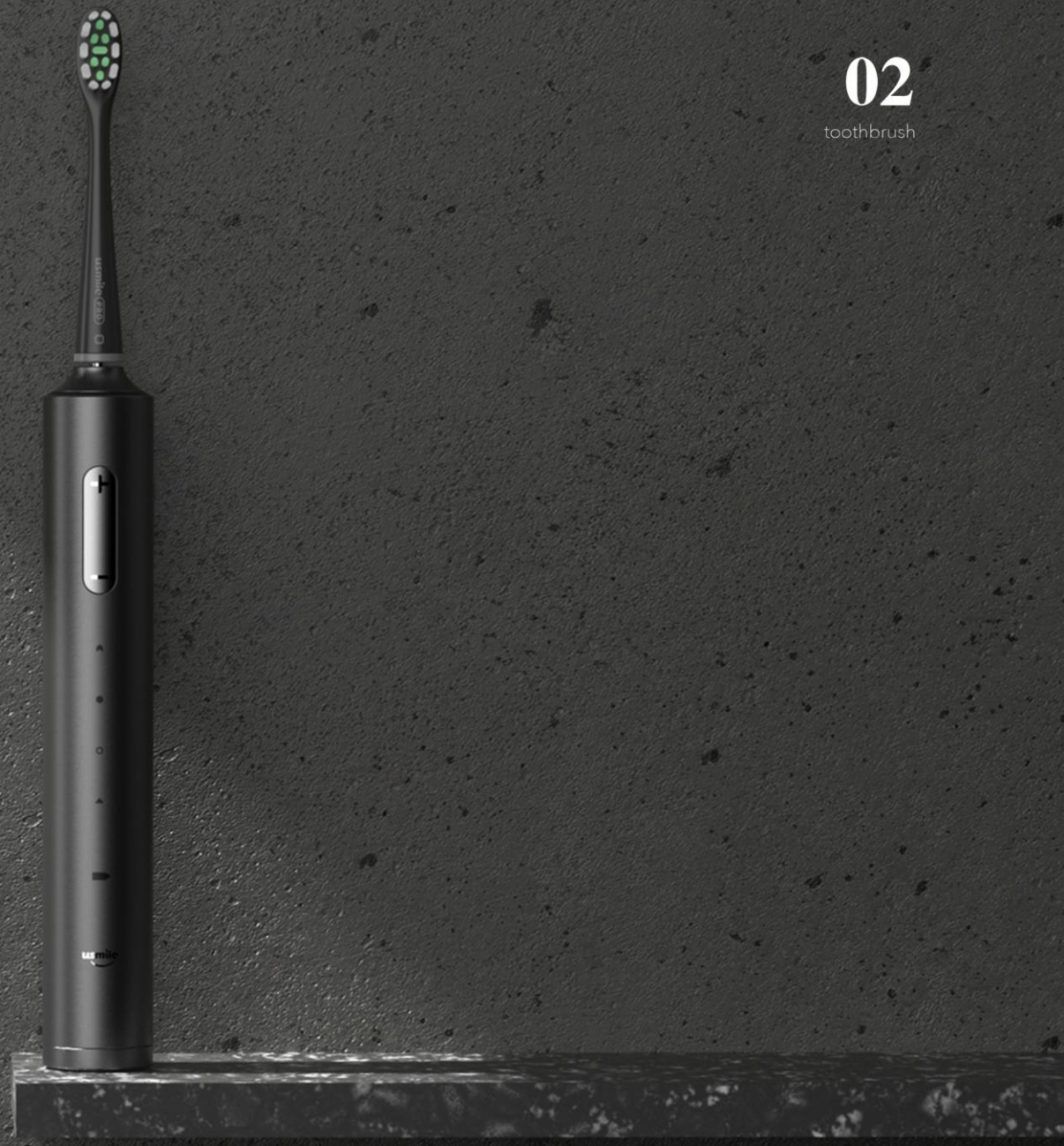


Toothbrush/M

(Aim to design a toothbrush for male user group)

02

toothbrush

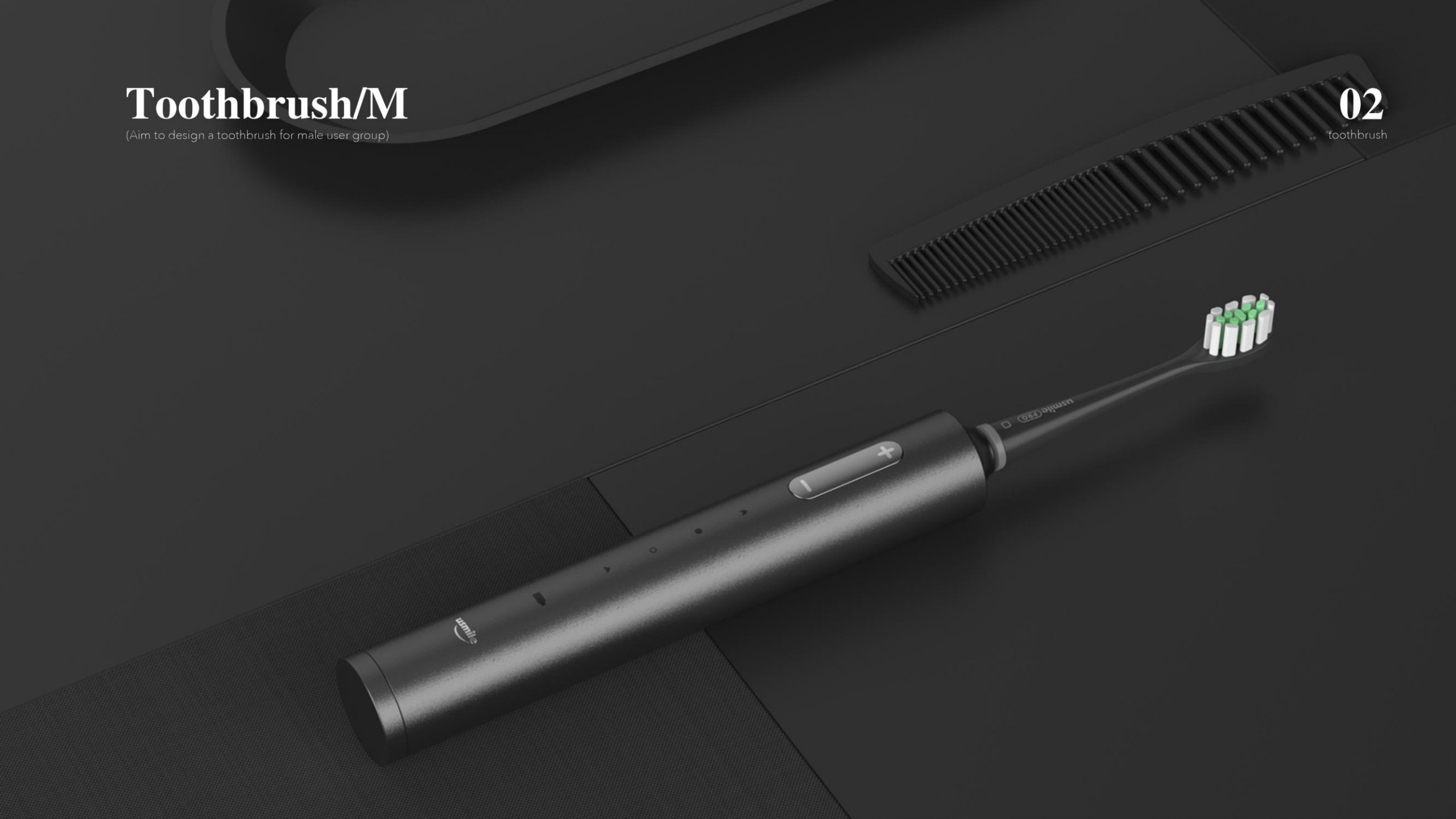


Toothbrush/M

(Aim to design a toothbrush for male user group)

02

toothbrush



Toothbrush/M

(Aim to design a toothbrush for male user group)

02

toothbrush







Facial Lifting Machine

(High-Frequency EMS Facial Lifting Machine)

Tri-Zone Premium Facial Lifting Device

1. Data-Driven User Profiling
The project began with user interviews and questionnaires to build a clear user persona. The goal was to identify the most ergonomic grip and determine which functional areas delivered the greatest comfort.

2. Strategic 3-Zone Functionality
Based on product management requirements, three distinct functional zones were defined. These zones were aligned with the user's natural facial contours to ensure the device was not only feature-rich, but also intuitive to use.

3. Market-Specific Aesthetic
To create a more premium look, an electroplated gold finish was selected. This bold visual choice was informed by beauty industry trends, where strong metallic tones and expressive detailing are often associated with high performance and luxury.

4. The Result
A high-end beauty instrument that balances complex multi-functionality with a premium aesthetic. It's a example of performance meeting luxury to drive consumer desire.



Facial Lifting Machine

(High-Frequency EMS Facial Lifting Machine)

Consumer view

Based on our survey of the five best-selling products, we analyzed user preferences regarding form and aesthetics.

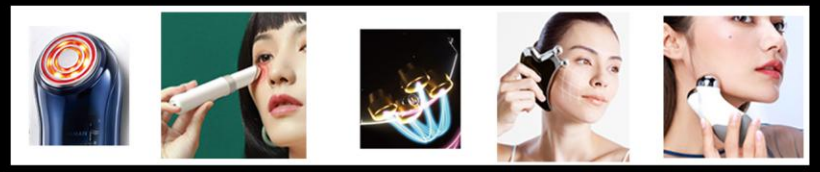
Questionnaire: view on shape

A B C D E



We conducted offline testing of various "functional heads" to identify the most ergonomic and comfortable for skin contact.

A B C D E



Aesthetics



Easy to use



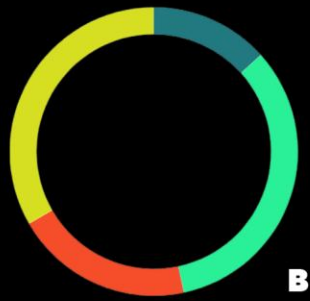
looks expensive



RF



EMS



EMS for eyes



Users tend to favor clean, minimalist forms that convey a sense of quality and refinement. Our research also shows a strong preference for metallic finishes, while a rounded functional head delivers a more comfortable user experience.

Facial Lifting Machine

(High-Frequency EMS Facial Lifting Machine)

Big picture

I conducted a visual analysis of top-selling beauty devices to uncover common design patterns and shared visual language within the category.

Sum of business









price →

Skin care product

By analyzing the skincare products our target users already use daily, we aligned the device's design language with existing premium aesthetic cues.



Consumer data indicates a strong preference for metallic finishes. We also observed a clear relationship between color palette and perceived value, with darker, richer tones consistently associated with a more premium and professional feel.

Facial Lifting Machine

(High-Frequency EMS Facial Lifting Machine)

04

Beauty



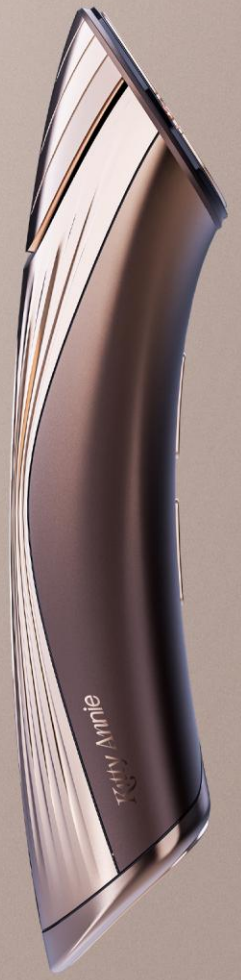
Facial Lifting Machine

(High-Frequency EMS Facial Lifting Machine)

04

Beauty







Facial Lifting Machine

(High-Frequency EMS Facial Lifting Machine)

04

Beauty





Package bag

(gift wrapping

05

bag

The Eco-Gift Wrap Dual-Purpose Packaging

1. The Sustainable Concept
Traditional disposable gift packaging was reimagined as a reusable everyday bag. This approach supports modern sustainability goals by reducing packaging waste.

2. Logical Space Optimization
Working within a compact form factor that could not accommodate a traditional paper manual, the instructions were integrated directly into the packaging as functional seals. This ensures users have the necessary information without additional printed materials.

3. Quality Assurance & Trust
Breaking the seal clearly indicates that the customer is the first to open the package, reinforcing product authenticity and trust.

4. The Result
An efficient, eco packaging system that eliminates the need for both a separate manual and a shopping bag — reducing cost and waste while strengthening user trust.



Package bag

(gift wrapping)

05

bag



**Users can use it in two ways,
hand-carried or as a messenger bag.**

Rocket toothbrush

(for 6 years old)



Grow-With-Me
Pediatric Electric Toothbrush

1. Balancing Playfulness and Professionalism
Designing for a six-year-old requires careful balance. The goal was to create a product that feels friendly and approachable for children, while maintaining a level of professionalism that reassures parents of its effectiveness.

2. Ergonomics & Safety
The grip patterns and motor skills of young children were studied to inform the design. The form was optimized for smaller hands, making it easy to hold and safely without adult assistance.

3. Emotional Design
Friendly visual elements were incorporated to transform a daily routine into a more engaging experience. By designing the toothbrush to feel like a companion rather than a clinical tool, resistance was reduced and healthy habits were encouraged.

4. The Result
A trusted oral care solution that parents feel confident purchasing — and that children are genuinely excited to use. It successfully bridges functional healthcare with playful engagement.

Little seahorse

(New packaging design for our dental floss stick)

New floss packaging

This packaging is designed to relate to our "Little Seahorse" dental floss stick.





Gift bag

(free bag)

08
bag

Viral Branding & Packaging

1. Customer Retention
This low-cost, portable pouch was designed as a gift-with-purchase to strengthen brand loyalty. By offering a quality ok item, it creates an additional touchpoint that encourages customers to choose our brand over competitors.

2. The Collectibility Factor
The pouch was released in a multi-color series to encourage users to collect different versions. This strategy is aim to drives repeat purchases and elevates a simple package into a accessory.

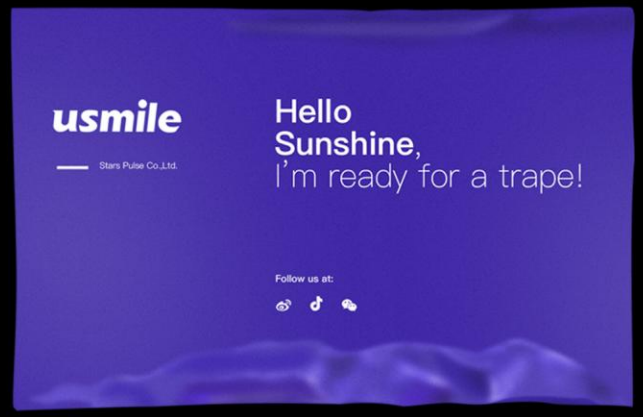
3. Social Media Integration
The design incorporates a bold, playful slogan intended to be highly shareable. Encourages users to post their unboxing experience on social media, generating exposure and increasing brand visibility.

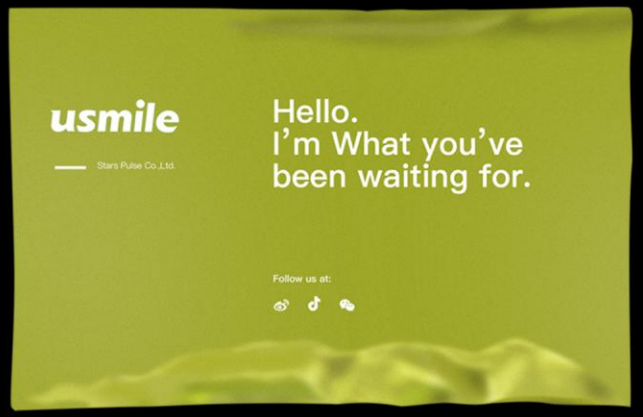
4. The Result
An high-impact marketing product that strengthens brand recognition and customer loyalty. Through engaging design, it helps convert first-time buyers into long-term advocates.



Gift bag

(free bag)













Colors design
and is on the market

(preview)

09







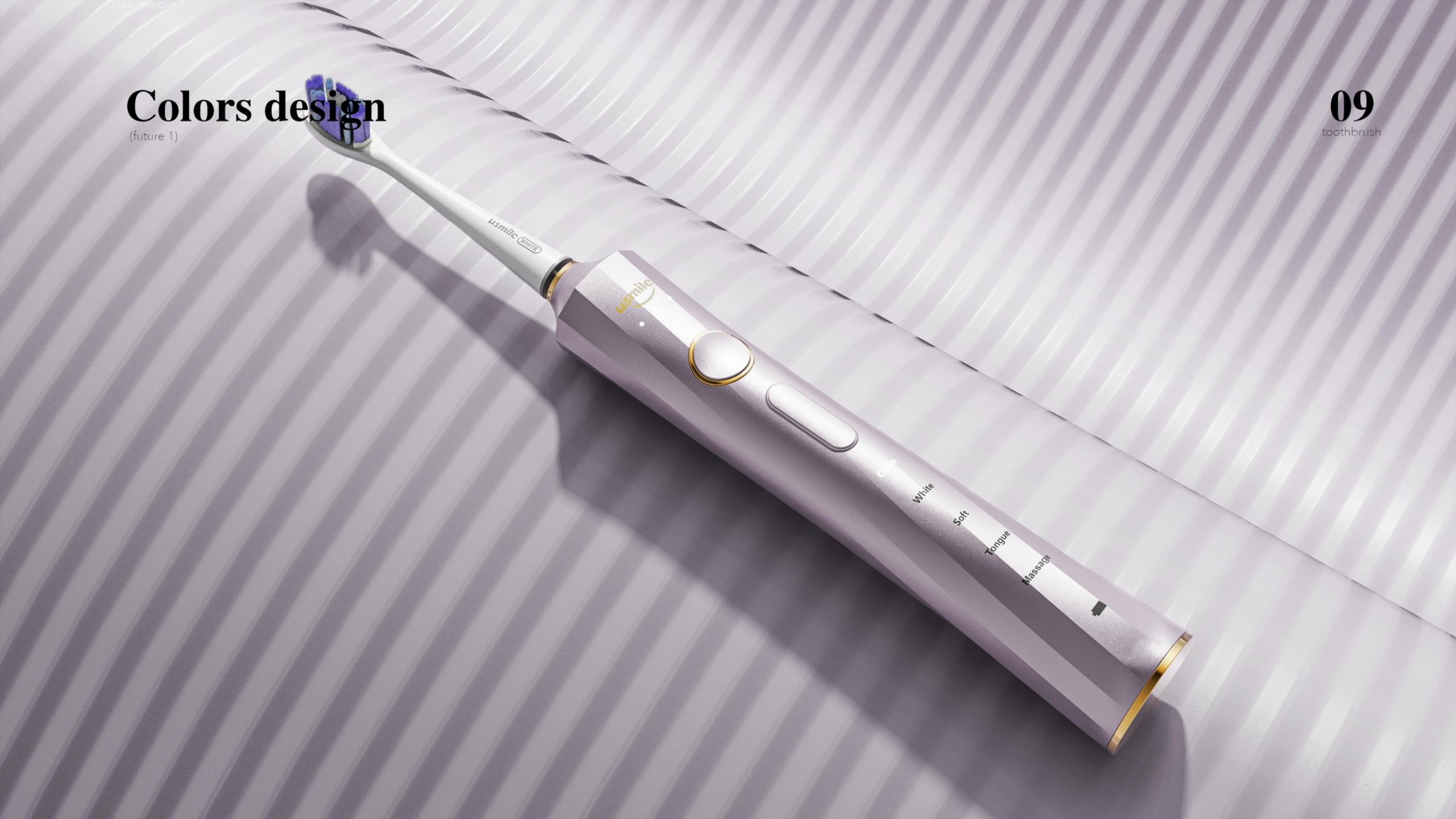


Colors design

(future 1)

09

toothbrush



Colors design

(future 1)

09

toothbrush

Future 1.

I studied how leading brands use color to separate their premium products from basic ones. Which is by using muted tones and metallic finishes.



Colors design

(future 1)

09

toothbrush

Future 1.

I studied how leading brands use color to separate their premium products from basic ones. Which is by using muted tones and metallic finishes.



I found that luxury products shift from high saturation to muted tones with more metallic textures to look expensive.

Colors design

(future 1)

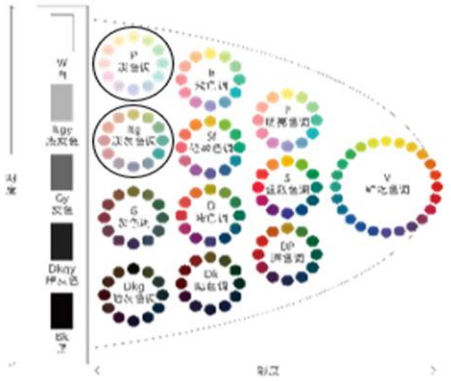
09

toothbrush

Combined with survey research conducted by a design agency.

桃美人 Pachyphytumcv	Pantone 2234 C	RAL 190-70-25	NCS S 3030-B50G
初晴 After The Rain	Pantone 7606 C	RAL 040-70-30	NCS S 2030-Y70R
丁香 Lilac	Pantone 2635 C	RAL 290-80-15	NCS S 1030-R60B
蓬莱 Penglai	Pantone 2230 C - 7707 C	RAL 190-50-40 - 220-40-40	NCS S 3050-B40G - 4050-B10G
冷仙 Icy Fairy	Pantone 2377 C	RAL 250-30-25	NCS S 6030-R90B
琉璃染 Coloured Dye	Pantone 2025 C	RAL 050-70-60	NCS S 0570-Y60R

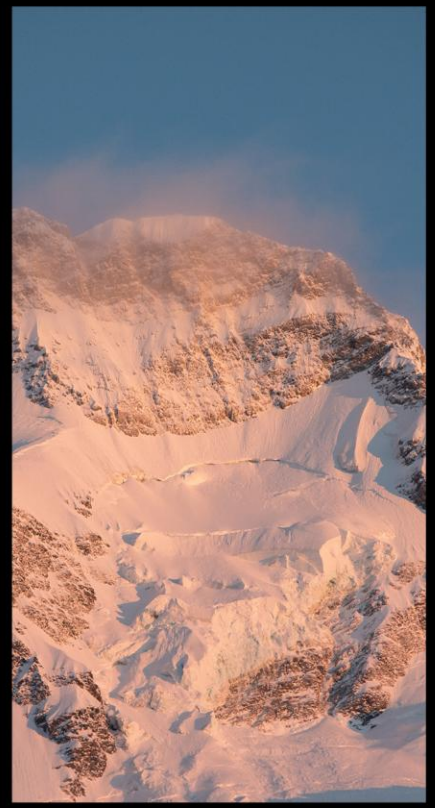
丝 Silk	Pantone Warm Grey 1 C	RAL 100-90-05	NCS S 1505-Y20R
银釉 Sliver Glaze	Pantone 877 C	RAL 820-M	NCS S 4502–B
雪国 Snow Land	Pantone 290 C	RAL 230-85-15	NCS S 0520-B
雨后桂花 Osmanthus	Pantone 7499 C	RAL 085-90-20	NCS S 0510-Y10R
银狐 Sliver Fox	Pantone 664 C	RAL 290-92-05	NCS S 0510-R60B
白桦 White Birch	Pantone 663 C	RAL 000-90-00	NCS S 0500-N



Colors design

(future 1)

Design Scenes comes out




Colors design

(future 1)

09

toothbrush

Out come









Colors design
(future 1)

On the market





Colors design

(future 1)

On the market





Colors design

(Hair remover)

it’s A hair remover



Colors design

(Hair remover)

09

Beauty

it’s A hair remover



Colors design

(Hair remover)

09

Beauty

it’s A hair remover



Water flosser

(Aim to design a water flosser that has it own topic)

Water flosser

1. Rapid-Cycle Development
I took this project from research to final 3D rendering in just 10 days. The brief focused on creating a visually striking product designed to appeal to female consumers and perform well on social media.

2. Cross-Category Inspiration
Inspired by design approaches outside the oral care category, I studied transparent electronics and unconventional geometry to move beyond traditional dental product aesthetics.

3. Visual Disruption Strategy
The flosser was developed to stand out instantly on social media. By positioning it as a lifestyle accessory rather than a purely medical tool, a bold and shareable visual identity was created.

4. The Result
A stylish, market-ready flosser that demonstrates how everyday functional products can also make a bold visual statement.



Water flosser

(Aim to design a water flosser that has it own topic)

10

water flosser



Water flosser

(Aim to design a water flosser that has it own topic)

10

water flosser



Water flosser

(Aim to design a water flosser that has it own topic)



Water flosser

(Aim to design a water flosser that has it own topic)

10

water flosser



BlU ‘Family’
(Compliance & Iteration)

Blu Next-Gen Series

1. Brand Iteration
Three new models were developed for blu, beginning with the cylindrical disposable.I kept the brand's original style but upgraded the design language.

2. Ergonomic Focus
User comfort was prioritized by refining the mouthpiece for improved lip feel. A consistent mouthpiece design across both the cylindrical and box models to make the whole product line feel like one family.

3. Innovative Compliance
To meet new European laws, I created a first of its kind rotating child lock. This clever mechanical solution allows the battery to be detached easily while keeping the product safe and legal.

4. Material Upgrade
A improved on the texture and surface treatment of the replaceable mouthpiece one. This makes the product feel higher quality while maintaining a clean look for the European market.



Webcam
(Design to Production)

Webcam with Wireless Lavalier Mic.

1. User-Informed Direction
User questionnaires explored preferences for webcam appearance and privacy features. These insights established a clear design direction, aligning the product with user expectations.

2. Clear Privacy Control
Feedback highlighted a preference for a flip-up privacy cover that visibly blocks the lens. This physical solution offers clear reassurance and makes privacy control intuitive.

3. Market-Informed Form
For an emerging brand, comparisons with competing webcams in the same price range helped identify a familiar and appealing aesthetic. Users favored a rounded, pill-shaped form, which became the foundation of the design.

4. From Insights to Launch
The final design brings together user preferences, a refined form, and visible privacy protection. Translating research insights to a solution.



Webcam
(Design to Production)

Webcam for Streamers

1. Designed for Game Streamers
Designed specifically for game streamers, the webcam complements the gaming setups and peripherals they use every day.

2. Research into Gaming Aesthetics
Research into gaming setups, mice, and keyboards revealed RGB lighting as a common element of gaming product design. This insight helped shape the webcam's visual direction.

3. A Distinctive Visual Identity
Although common across gaming peripherals, RGB lighting was rarely seen in the webcams reviewed during the research. Incorporating it into the design gave the webcam a distinctive identity that connects with the wider gaming setup.

4. Positive User Feedback
The design received highly positive feedback from users, suggesting that its gaming-inspired appearance resonated with the target audience.



Webcam
(Design to Production)

Webcam with a speaker

1. Making Sound Visible
For a webcam with a built-in speaker, the key challenge was communicating its audio capability at a glance. With limited brand recognition, the design needed to help potential buyers immediately recognize both functions.

2. Learning from User Feedback
Several rounds of design comparisons revealed that users did not readily identify the built-in speaker. This highlighted the need for a stronger visual cue to communicate the product's audio function.

3. Drawing on Familiar Cues
JBL's RGB-lit Bluetooth speakers inspired a new direction. Incorporating similar lighting cues offered a way to draw on familiar associations with audio products and make the webcam's speaker function more apparent.

4. Positive User Response
The revised design received positive feedback in follow-up questionnaires. The response supported the use of familiar visual cues to communicate an unfamiliar product concept.



Monitor
(Design to Production)

Gentle Care

1. A Flagship Monitor
Designed as the company's most premium baby monitor display, the project aimed to combine a welcoming appearance with intuitive everyday use.

2. A Softer Design Language
Hands-on evaluation of the company's existing products and leading competitors revealed a largely utilitarian approach to design. A softer, more approachable aesthetic was chosen to better suit a product made for families with babies.

3. More Accessible Controls
Many of the monitors reviewed relied on touch controls or small buttons that were difficult to use. The new design addressed these issues with larger physical buttons, making key functions easier to access and operate.

4. Positive User Feedback
The combination of a gentler visual language and practical controls received positive feedback from users, reflecting the value of addressing both emotional and functional needs.



Camera
(Design to Production)

NightNest

1. Clearer Views After Dark
The main challenge was capturing a clear view of the baby inside a dark vehicle. The existing infrared illumination was insufficient, making additional lighting necessary for nighttime use.

2. A Gentler Lighting Approach
Visible light offered a way to improve visibility, but needed careful placement to avoid shining directly into the baby's face.

3. Integrated Indirect Lighting
The supplementary light was concealed beneath the camera, allowing illumination to spread softly around the housing. This indirect approach was designed to improve the camera's view while keeping the light source out of the baby's direct line of sight.

4. Designed for Nighttime Care
The design balances nighttime visibility with a more considerate lighting experience, helping parents check on their baby in low-light conditions.



Camera
(Design to Production)

Wireless Baby Camera with Temperature Monitoring

1. Wireless Convenience
Wireless connectivity offers greater flexibility in everyday use, making it easier to position the camera for a clear view of the baby.

2. Beyond the Camera
Temperature monitoring adds another layer of information, helping caregivers stay aware of the baby's surroundings alongside the live video.

3. Designed for Reassurance
The concept brings together visual and environmental information to help caregivers feel better informed when checking on their baby.



Dash cam

(Design to Production)

RoadFrame

1. Controls You Can Feel
The side controls use contrasting shapes and textures to make them easier to distinguish by touch. A raised, textured button stands apart from the recessed buttons below, while clear grouping reduces reliance on small printed labels.

2. Ventilation Around the Lens
Continuous recording and exposure to sunlight make thermal management an important design consideration. Radial openings around the lens provide space for ventilation while creating an orderly front layout, turning a functional requirement into a recognizable design feature.

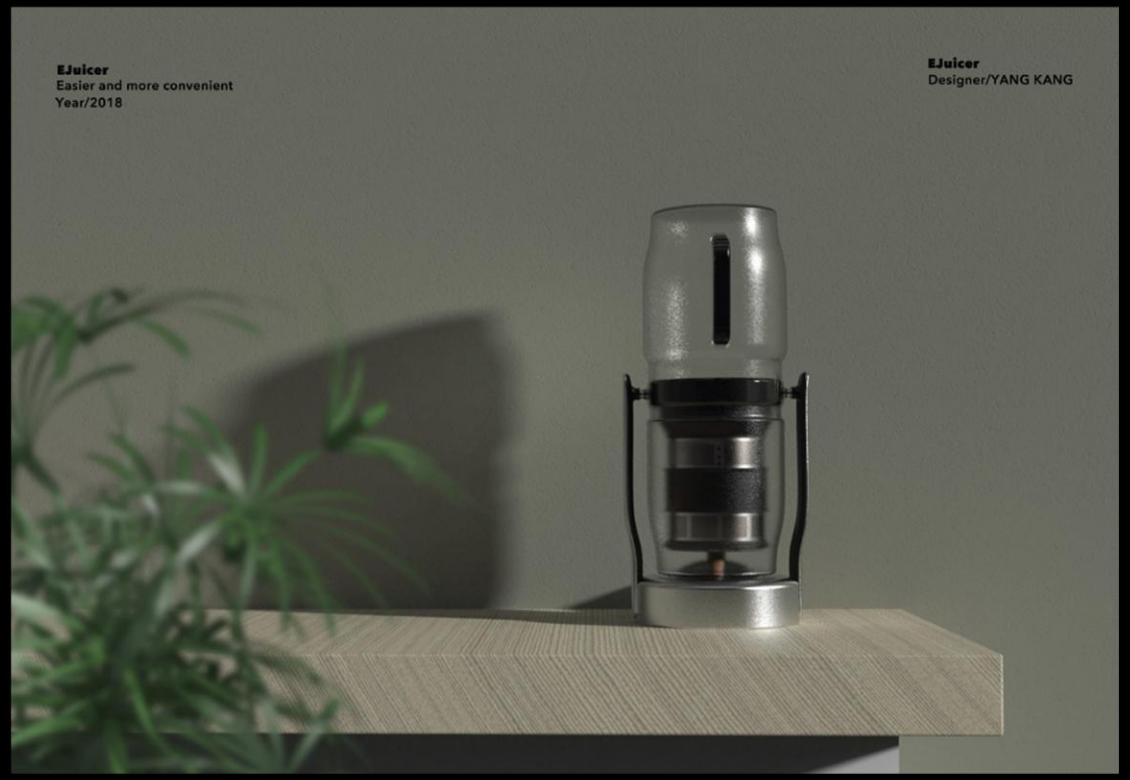
3. A Quiet Presence in the Cabin

Positioned near the windshield, the dash cam needs to sit comfortably within the vehicle interior. A dark matte body, rounded edges, and restrained metallic accents convey precision while keeping reflections and visual distraction to a minimum.



Passion about design

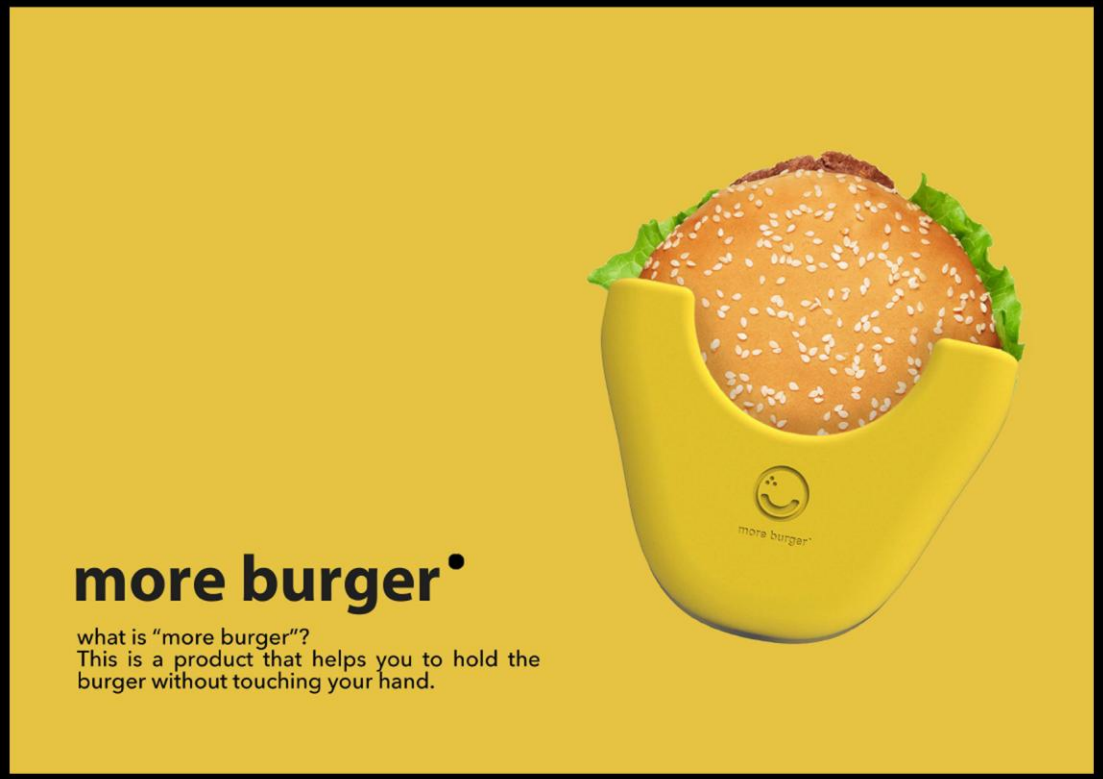
(Some project from my time as a student)

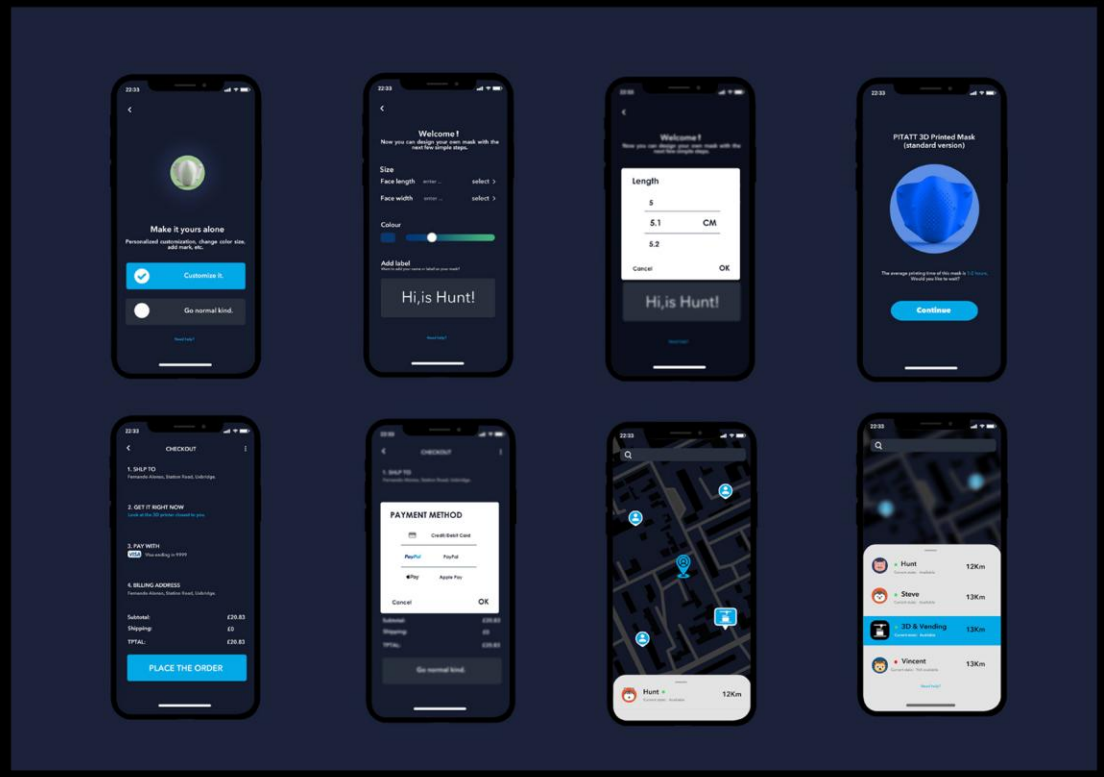




Passion about design

(Some project from my time as a student)





Thank you

(Hope to see you soon)

the end.